IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

SEMICONDUCTOR DEVICES

Thursday, 4 December 2025, 10:00 AM

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer question ONE. Answer any TWO of the remaining questions.

Calculators are permitted.

Formula sheet will be supplied with the exam paper.

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : K. Fobelets

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Constants

Permittivity of free space:	$\varepsilon_o = 8.85 \times 10^{-12}$ F/m
Permeability of free space:	$\mu_o = 4\pi \times 10^{-7}$ H/m
Intrinsic carrier concentration in Si:	$n_i = 1.45 \times 10^{10}$ cm^{-3} at $T = 300$ K
Dielectric constant of Si:	$\varepsilon_{Si} = 11$
Dielectric constant of SiO_2:	$\varepsilon_{ox} = 4$
Thermal voltage:	$kT/e = 0.026$ V at $T = 300$ K
Charge of an electron:	$e = 1.6 \times 10^{-19}$ C
Planck's constant:	$h = 6.63 \times 10^{-34}$ Js
Boltzmann constant:	$k = 1.38 \times 10^{-23}$ J/K
Band gap of Si:	$E_G = 1.12$ eV at $T = 300$ K
Electron affinity of Si:	$\chi = 4.05$ eV at $T = 300$ K
Effective density of states of Si:	$N_C = 3.2 \times 10^{19}$ cm^{-3} at $T = 300$ K $N_V = 1.8 \times 10^{19}$ cm^{-3} at $T = 300$ K

Formulae

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2}+V(x)\psi(x)=E\psi(x)$$ Schrödinger's equation in one dimension

$$f(E)=\frac{1}{1+\exp\left(\frac{E-E_f}{kT}\right)}$$ Fermi distribution

$$m_e^*=\frac{\hbar^2}{d^2E(k)/dk^2}$$ Effective mass

$$n=\frac{1}{\sqrt{2}\hbar^3}\left(\frac{m_e^*kT}{\pi}\right)^{3/2}e^{\left(-\frac{E_c-E_F}{kT}\right)}=N_ce^{\left(-\frac{E_c-E_F}{kT}\right)}$$ Concentration of electrons

$$p=\frac{1}{\sqrt{2}\hbar^3}\left(\frac{m_h^*kT}{\pi}\right)^{3/2}e^{\left(-\frac{E_F-E_v}{kT}\right)}=N_ve^{\left(-\frac{E_F-E_v}{kT}\right)}$$ Concentration of holes

$$\left.\begin{aligned}J_n(x)&=e\mu_n n(x)E(x)+eD_n\frac{dn(x)}{dx}\\J_p(x)&=e\mu_p p(x)E(x)-eD_p\frac{dp(x)}{dx}\end{aligned}\right\}$$ Drift and diffusion current densities in a semiconductor

$$I_{DS}=\frac{\mu C_{ox}W}{L}\left((V_{GS}-V_{th})V_{DS}-\frac{V_{DS}^2}{2}\right)$$ Current in a MOSFET

$$\left.\begin{aligned}J_n&=\frac{eD_n n_{p_0}}{L_n}\left(e^{\frac{eV}{kT}}-1\right)\\J_p&=\frac{eD_p p_{n_0}}{L_p}\left(e^{\frac{eV}{kT}}-1\right)\end{aligned}\right\}$$ Diffusion current densities in a pn-junction

$$V_0=\frac{kT}{e}\ln\left(\frac{N_AN_D}{n_i^2}\right)$$ Built-in voltage

$$c=c_0\exp\left(\frac{eV}{kT}\right)\text{ with }\begin{cases}c=p_n\text{ or }n_p\\c_0\text{ bulk minority carrier concentration}\end{cases}$$ Minority carrier injection under bias V

$$D=\frac{kT}{e}\mu$$ Einstein relation

$$w_n=\sqrt{\frac{2\varepsilon_0\varepsilon_r}{q}\frac{N_A}{N_AN_D+N_D^2}(V_0-V)}$$

$$w_p=\sqrt{\frac{2\varepsilon_0\varepsilon_r}{q}\frac{N_D}{N_AN_D+N_A^2}(V_0-V)}$$ Depletion width in the n and p-type region, under bias V

$$\delta c = \Delta c \exp\left(\frac{-x}{L}\right) \text{ with } \begin{cases} \delta c = \delta p_n \text{ or } \delta n_p \\ \Delta c \text{ the excess carrier concentration} \\ \text{at the edge of the depletion region} \end{cases}$$

Excess carrier concentration as a function of distance when recombination occurs – long layer approximation.

$$L = \sqrt{D\tau}$$

Diffusion length

$$C_{diff} = \frac{e}{kT} I\tau$$

Diffusion capacitance

$$i(t) = \frac{Q(t)}{\tau} + \frac{dQ(t)}{dt}$$

Time variation of current and charge

$$\frac{dE}{dx} = \frac{\rho(x)}{\varepsilon}$$

Poisson equation in 1 dimension

$$n = N_C \, exp\left(\frac{E_F - E_C}{kT}\right)$$

$$p = N_V \, exp\left(\frac{E_V - E_F}{kT}\right)$$

Carrier concentrations

1. [This question carries 40% marks]

a) The current density of electrons in a semiconductor is given by:

$$J_n(x) = e\mu_n n(x)E(x) + eD_n \frac{dn(x)}{dx}$$

Briefly explain the physical processes which give rise to the two terms on the right-hand side of the equation.

[4]

b) By considering material work functions and drawing energy band diagrams, explain how a Schottky contact forms on a *p*-type semiconductor.

[4]

c) Consider a semiconductor volume of infinitesimal width Δx and constant cross-sectional area A, with an electric field $\boldsymbol{E}$ applied along the x-axis. By using Gauss's Law, $\int_S \boldsymbol{E}.\mathrm{d}\boldsymbol{A} = \frac{\rho(x)}{\epsilon}$, derive Poisson's Equation along the x-axis, for the semiconductor. Here, $\mathrm{d}\boldsymbol{A}$ the normal vector to infinitesimal area $\mathrm{d}A$, ρ(x) the charge density and ε the permittivity within the volume. Your answer should state the different possible contributions to the charge density.

[6]

d) Sketch the band diagram (E_C, E_V, E_F, E_G), from the emitter to the collector, for an n^+pn bipolar junction transistor (BJT) with:

i) The BJT with zero applied bias.

[4]

ii) The BJT in cut-off.

[4]

iii) The BJT in forward active mode.

[4]

Question continues on the next page

e) The output characteristics of an enhancement mode *n*-channel MOSFET at room temperature are shown in Fig. 1.1. The MOSFET is assumed to operate as a 'long-channel transistor. The dashed line is the locus where $V_{DS} = V_{GS} - V_{th}$. Answer the following questions for this MOSFET.

i) Give the gate voltage V_{GSi} in ascending order (use <).

[2]

ii) Give the expression for the current I_{DS} for low V_{DS} in region 1.

[4]

iii) Give the expression for the current I_{DS} for high $V_{DS} > V_{GS} - V_{th}$.

[4]

iv) Give the expression for the current I_{DS} as a function of V_{DS} for the dashed line.

[4]

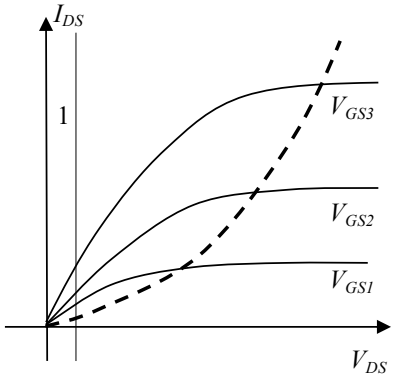


Figure 1.1

2. [This question carries 30% marks]

a)

Consider the energy band diagram, in a silicon semiconductor at temperature T = 300 K, given in Fig. 2.1.

i. Calculate the density of majority carriers in the semiconductor.

[4]

ii. Calculate the value of E_i (in eV) relative to E_c, at temperature T = 300 K.

[8]

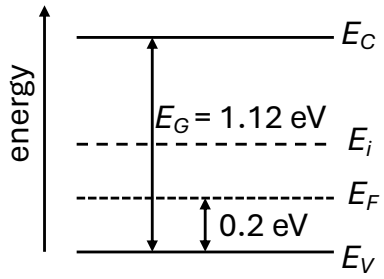


Figure 2.1

b)

A p^+n diode is biased using a voltage V and a series resistive load resistor R, as shown in Fig. 2.2. Switch 'S' is used to switch the diode 'on' or 'off'.

(i) Switch 'S' is initially open. The diode is then switched 'off' to 'on' by closing 'S'. Derive the expression for, and sketch the time variation of, the excess hole concentration $Q_p(t)$ in the n region. State any assumptions you make.

[6]

(ii) The diode is to be switched from 'on' to reverse bias, by applying a voltage -V at time t = 0. Draw the excess hole distribution vs. length in the n region, at $t = 0^-$, $t = 0^+$, and at t corresponding to the final, steady state.

[6]

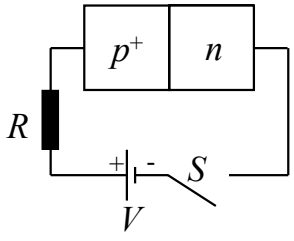


Figure 2.2

(c) A p^+np BJT, connected in common-emitter mode, is biased in the cut-off region. The BJT is then switched 'on' by biasing it in the forward active region using the emitter-base voltage. By considering the BJT biasing circuit, describe briefly the phenomenon of oversaturation in the BJT as it is switched 'on'.

[6]

3. [This question carries 30% marks]

a)

Figure 3.1 shows a sketch of the minority carrier concentration along the width of the base layer of a p^+np BJT. What is the correct label for the vertical axis (δp_n or δn_p)? Hence, by using the sketch in Fig. 3.1 and expressions for diffusion current, explain how the recombination current in the base can be calculated.

[8]

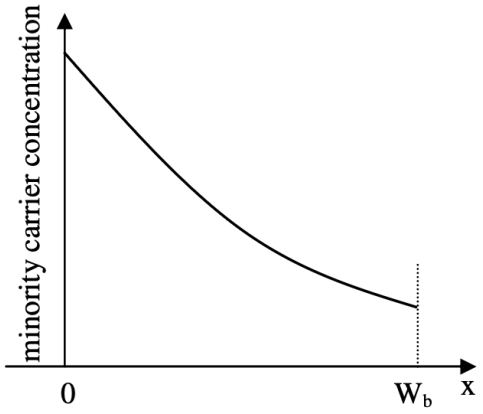


Figure 3.1

b)

Consider an n^+pn BJT with emitter efficiency $\gamma = 1$. Assume that the BJT base is short, although recombination takes place. If the excess minority carrier charge in the base is given by Q_b, give expressions for the base current I_b and the collector current I_c as a function of Q_b. Define the parameters you use.

[8]

c)

Calculate the current gain β in an n^+pn BJT in forward active mode, with $\gamma = 1$. Material parameters for the BJT are listed in Table 3.1. You may neglect the influence of depletion only for the forward biased junction. The applied voltages are $V_{EB} = 0.5$ V and $V_{BC} = -1$ V and temperature $T = 300$ K. You may approximate the transit time using the expression $\tau_t = \frac{W_B^2}{2D_n}$.

[14]

Material parameters	emitter	base	collector
Doping density (cm^{-3})	10^{20}	10^{17}	5×10^{16}
Region length (μm)	0.1	1	500
Diffusion constant ($cm^2\ s^{-1}$)	2	6	20
Minority carrier lifetime (s)	1×10^{-10}	1×10^{-6}	1×10^{-2}
Cross sectional area (μm^2)	100	100	100

Table 3.1

4. [This question carries 30% marks]

Consider an *n*-channel enhancement mode Si MOSFET with gate length L, gate width W, gate oxide thickness t_{ox} and gate capacitance per unit area C_{ox}. The donor doping concentration, per unit volume, in the source and drain regions is N_D. The acceptor concentration, per unit volume, in the bulk is N_A. A drain-source voltage V_{ds} and gate-source voltages V_{gs} is applied to the MOSFET, and the source is grounded. The MOSFET threshold voltage in V_{th}.

a)

(i) Is the threshold voltage V_{th} in this MOSFET greater than, or less than, zero?

[2]

(ii) The MOSFET channel length lies along x-axis. Find an expression for the inversion layer charge per unit area, Q_n, as a function of the voltage drop $V_x(x)$ along the channel. State any approximation you make.

[8]

b) Using your results from (a), write down expressions for Q_n at the source and drain ends of the channel region at 'pinch-off'.

[4]

c) Hence, by considering the drift current in the channel and the electron mobility μ_e, derive the following expression for the channel current I:

$$I = C_{ox}\mu_e \frac{W}{L}\left((V_{gs} - V_{th})V_{ds} - \frac{V_{ds}^2}{2}\right)$$

[16]

[END]